CS445 / SE463 / CS645

Software Requirements Specification and Analysis

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Deliverable 6: Cost Estimation, Risk Analysis, and LTL Constraints

Group Name: The Reservists
System Name: Carrel
Members: Kevin Wang, Chris Wu, Rui Lin
Course: CS445 / SE463 / CS645

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1 Overview and Assumptions

This deliverable continues the same system used in Deliverables 4 and 5. **Carrel** is a campus study-space booking system that allows students to search for spaces, reserve a space, check in, cancel future bookings, file disputes, and receive booking-related notices. Administrators can manage the study-space inventory, booking rules, maintenance closures, disputes, overrides, and usage analytics.

The assignment requires three major parts:

1) a preliminary effort estimate using function points, code size from function points, and COCOMO;
2) a risk analysis using a risk consequences table and a risk countermeasures table; and
3) ten quality/long-term behavioral attributes written once in English and once in Linear Temporal Logic (LTL).

The estimates below are preliminary requirements-time estimates. They are not commitments. The calculation is intentionally shown step by step so that every number in the final estimate can be traced back to an assumption or a formula.

2 Preliminary Effort Estimate

2.1 Scope of the Function Point Estimate

The function point estimate focuses on top-priority use cases from the previous deliverables. These are the use cases that directly support the main student booking lifecycle and the administrator controls needed to keep that lifecycle safe and reliable.

Table 1: Top-priority use cases used for the effort estimate

Use Case	Name	Reason for Inclusion
UC14	Authenticate User	All booking and administration behavior depends on valid user identity and role.
UC1	Search Available Study Spaces	This is the entry point to the student booking flow.
UC2	View Study Space Details	Students need space details before reserving.
UC3	Reserve Study Space	This is the core value-producing use case.
UC4	View My Bookings	Students need to monitor upcoming and active bookings.
UC5	Check In to Booking	Required to prevent no-show misuse.
UC6	Cancel Booking	Required to return unused space to the available pool.
UC7	File Booking Dispute	Required to challenge no-show or room conflict outcomes.
UC8	Manage Study Space Inventory	Administrators must keep bookable space data accurate.
UC9	Configure Booking Rules	Administrators must configure quotas, grace windows, slot lengths, and no-show rules.

Use Case	Name	Reason for Inclusion
UC10	Manage Maintenance Closures	Administrators must block unavailable spaces.
UC11	Review and Resolve Disputes	Administrators need to process submitted disputes.
UC12	Apply Administrative Override	Administrators need controlled override capability for exceptional cases.
UC13	View Usage Analytics	Useful for monitoring peak demand and under-used spaces.
UC15	Send Notification	Needed for confirmations, cancellations, no-show notices, and dispute updates.
UC16	Create Calendar Invitation	Needed to integrate confirmed bookings with user calendars.
UC17	Release No-show Booking	Needed to automatically release unused bookings after the grace window.

2.2 Function Point Method Used

The basic function point model used here is:

$$FP = a_1 EI + a_2 EO + a_3 EQ + a_4 EIF + a_5 ILF$$

where:

- **EI** means External Inputs: user inputs or input events that add, change, or trigger processing.
- **EO** means External Outputs: generated outputs such as confirmation messages, reports, and notices.
- **EQ** means External Inquiries: request/response queries that do not mainly change system state.
- **EIF** means External Interface Files: external systems or external data sources used by Carrel.
- **ILF** means Internal Logical Files: internal logical data groups maintained by Carrel.

The weights are the standard course weights used in the lecture slides:

Table 2: Function point weights

Function Type	Low	Average	High
External Input (EI)	3	4	6
External Output (EO)	4	5	7
External Inquiry (EQ)	3	4	6
External Interface File (EIF)	5	7	10
Logical Internal File (ILF)	7	10	15

2.3 External Inputs

External Inputs are events that enter data into Carrel or trigger a state-changing operation. For example, reserving a study space creates a booking, and checking in changes a booking's lifecycle state. The following table lists each counted input and the reason for its complexity.

Table 3: External Input count

No.	External Input	Complexity	Weight	Reasoning
EI-1	Sign in / authenticate user	Average	4	Uses identity service and role lookup, but does not create many internal objects.
EI-2	Submit reservation request	High	6	Checks availability, booking rules, conflicts, closures, and creates booking-related objects.
EI-3	Submit check-in code or QR scan	Average	4	Validates code and changes booking/check-in state.
EI-4	Cancel a future booking	Average	4	Validates ownership and timing, creates cancellation record, updates booking.
EI-5	File a booking dispute	Average	4	Validates booking ownership and creates a dispute plus notification.
EI-6	Configure booking rules	High	6	Updates policy/rule data that affects many future reservations.
EI-7	Create or update maintenance closure	Average	4	Updates closure data and checks affected spaces.
EI-8	Apply administrative override	High	6	Can override booking, cancellation, dispute, or enforcement outcome.
EI-9	Resolve submitted dispute	Average	4	Updates dispute status and sends result notice.
EI-10	Trigger no-show release by system time	Average	4	Scheduled input event that releases unclaimed bookings and creates strike data.
EI-11	Add/edit/retire study space inventory	High	6	Updates space catalog, capacities, amenities, and accessibility data.
	Total EI contribution		**52**	$7 \times 4 + 4 \times 6 = 28 + 24 = 52$

2.4 External Outputs

External Outputs are outputs generated by Carrel, including confirmation screens, notices, reports, and error messages.

Table 4: External Output count

No.	External Output	Complexity	Weight	Reasoning
EO-1	Booking confirmation screen/message	Average	5	Confirms reservation and shows booking time and space.
EO-2	Reservation error output	Average	5	Explains rule violations, conflicts, or unavailable spaces.
EO-3	Check-in success/failure output	Average	5	Confirms check-in or explains invalid/expired code.
EO-4	Cancellation confirmation output	Average	5	Confirms cancellation and returned availability.
EO-5	Dispute submission/update output	Average	5	Confirms dispute submission and later resolution status.
EO-6	No-show notice output	Average	5	Notifies student of released booking and strike record.
EO-7	Booking-rule save/error output	Average	5	Reports saved rule changes or validation errors.
EO-8	Usage analytics report	High	7	Summarizes no-show rates, peak demand, and under-used spaces.
EO-9	Maintenance closure conflict output	Average	5	Reports whether closure affects active bookings.
EO-10	Calendar invite output	Average	5	Creates external calendar event data after booking confirmation.
	Total EO contribution		**52**	$9 \times 5 + 1 \times 7 = 45 + 7 = 52$

2.5 External Inquiries

External Inquiries are request/response functions where the main purpose is to retrieve or display information rather than create or change records.

Table 5: External Inquiry count

No.	External Inquiry	Complexity	Weight	Reasoning
EQ-1	Search available study spaces	High	6	Uses filters, availability, rules, and closure data.
EQ-2	View selected study space details	Average	4	Displays location, capacity, amenities, and notes.
EQ-3	View my bookings	Average	4	Retrieves upcoming, active, and relevant past bookings.
EQ-4	View usage analytics dashboard	High	6	Aggregates usage patterns, no-show rates, and demand data.
EQ-5	View submitted disputes	Average	4	Retrieves dispute list and details for admin review.
EQ-6	View maintenance closure schedule	Average	4	Retrieves closures by space and time interval.

No.	External Inquiry	Complexity	Weight	Reasoning
EQ-7	View study-space inventory	Average	4	Retrieves campus/building/floor/space catalog.
	Total EQ contribution		**32**	$5 \times 4 + 2 \times 6 = 20 + 12 = 32$

2.6 External Interface Files

External Interface Files are external systems that Carrel reads from, writes to, or depends on.

Table 6: External Interface File count

No.	EIF	Complexity	Weight	Reasoning
EIF-1	University Identity Service	Average	7	Provides authentication and role verification.
EIF-2	Calendar Service	Average	7	Receives booking event data and returns calendar synchronization status.
EIF-3	Notification Service	Average	7	Sends confirmations, reminders, no-show notices, and dispute updates.
	Total EIF contribution		**21**	$3 \times 7 = 21$

2.7 Internal Logical Files

Internal Logical Files are logical groups of data that Carrel maintains. These do not necessarily map one-to-one to database tables; they are logical data groups from the requirements perspective.

Table 7: Internal Logical File count

No.	Internal Logical File	Complexity	Weight	Reasoning
ILF-1	User, Student, Administrator profiles	Low	7	Stores identity-linked roles and account states.
ILF-2	Study-space catalog	Average	10	Includes campus, building, floor, space, amenity, capacity, and accessibility fields.
ILF-3	Booking lifecycle records	High	15	Central object with status, timing, check-in deadline, cancellation, release, and links.
ILF-4	CheckInCode and CheckIn records	Average	10	Stores generated codes, usage status, and physical presence confirmation.
ILF-5	Cancellation records	Low	7	Stores reason, timestamp, and booking relationship.

No.	Internal Logical File	Complexity	Weight	Reasoning
ILF-6	NoShowStrike records	Low	7	Stores strike state, expiration, active flag, and student link.
ILF-7	Dispute records	Average	10	Stores reason, status, resolution, and related booking/no-show.
ILF-8	BookingPolicy and BookingRule records	Average	10	Stores quotas, advance windows, slot lengths, grace periods, and no-show thresholds.
ILF-9	MaintenanceClosure records	Average	10	Stores closure intervals, status, affected study space, and reason.
ILF-10	Notification records	Low	7	Stores recipient, type, subject, and delivery status.
ILF-11	CalendarInvite records	Low	7	Stores invitation id, start/end, and cancellation link.
ILF-12	AdministrativeOverride records	Low	7	Stores override type, rationale, administrator, and applied timestamp.
	Total ILF contribution		**107**	$6\times7+5\times10+1\times15=42+50+15=107$

2.8 Unadjusted Function Point Total

The total unadjusted function point estimate is the sum of the five function type contributions.

$$\begin{aligned} FP &= EI + EO + EQ + EIF + ILF \\ &= 52 + 52 + 32 + 21 + 107 \\ &= 264 \end{aligned}$$

Table 8: Function point summary

Function Type	Count Logic	Contribution	Percent of FP
External Inputs	$7\times4+4\times6$	52	$52/264=19.7\%$
External Outputs	$9\times5+1\times7$	52	$52/264=19.7\%$
External Inquiries	$5\times4+2\times6$	32	$32/264=12.1\%$
External Interface Files	3×7	21	$21/264=8.0\%$
Internal Logical Files	$6\times7+5\times10+1\times15$	107	$107/264=40.5\%$
Total		**264**	**100.0%**

2.9 Estimating Code Size from Function Points

The lecture table provides statements of source code per unadjusted function point for several programming languages. Carrel would most likely be implemented with a modern web stack such as Type-

Script/React/Node and a database. Because the course table does not list TypeScript directly, this estimate uses the C++ value from the course table, **29 SLOC/UFP**, as a high-level language proxy. This keeps the calculation tied to the course method and avoids importing a different external table.

Formula

$$\text{Estimated SLOC} = FP \times \text{SLOC per UFP}$$

Calculation

$$\begin{aligned}\text{Estimated SLOC} &= 264 \times 29 \\ &= 7{,}656 \text{ SLOC}\end{aligned}$$

$$\text{Estimated KLOC} = \frac{7{,}656}{1{,}000} = 7.656 \text{ KLOC}$$

Table 9: Code size estimate from function points

Item	Value
Function points	264 UFP
Language proxy	C++ / high-level web implementation proxy
SLOC per UFP	29
Estimated SLOC	$264 \times 29 = 7{,}656$
Estimated KLOC	$7{,}656/1{,}000 = 7.656$

2.10 COCOMO Effort Estimate

The COCOMO effort equation is:

$$E = a \times KLOC^b \times X$$

where E is effort in person-months, $KLOC$ is the estimated project size in thousands of source lines of code, a and b depend on the project mode, and X is a product of project attribute multipliers.

2.10.1 Selecting the COCOMO Mode

The estimated size is 7.656 KLOC, which is less than 50 KLOC. The team project is also a relatively small application with familiar web/database patterns. Therefore, the **organic** mode is selected.

Table 10: COCOMO mode selection

Mode	Size Range	a, b	Decision
Organic	< 50 KLOC	$a = 2.4$, $b = 1.05$	Selected
Semi-detached	< 300 KLOC	$a = 3.0$, $b = 1.12$	Not selected; too large for this estimate
Embedded	tightly constrained systems	$a = 3.6$, $b = 1.20$	Not selected; Carrel is not embedded software

2.10.2 Project Attribute Multiplier X

The course equation allows an attribute multiplier X. We use a small multiplier above 1.0 because Carrel is not just a local program: it has authentication, a database, concurrency-sensitive booking behavior, and external services. The individual factors are documented below.

Table 11: Project attribute multiplier calculation

Factor	**Multiplier**	**Reasoning**
Reliability / integrity sensitivity	1.10	Booking conflicts, no-show release, and role enforcement require careful testing.
External integration complexity	1.05	Carrel depends on identity, notification, and calendar services.
Modern web tools and frameworks	1.00	Frameworks help productivity, so no extra penalty is added here.
Combined X	$1.10 \times 1.05 \times 1.00 = 1.155$	Rounded only in final display; calculations use 1.155.

2.10.3 Effort Calculation

Using $a = 2.4$, $b = 1.05$, $KLOC = 7.656$, and $X = 1.155$:

$$E = 2.4 \times 7.656^{1.05} \times 1.155$$

First compute the exponent term:

$$\begin{aligned} 7.656^{1.05} &= e^{1.05 \ln(7.656)} \\ \ln(7.656) &\approx 2.0355 \\ 1.05 \ln(7.656) &\approx 1.05 \times 2.0355 = 2.1373 \\ e^{2.1373} &\approx 8.477 \\ \therefore 7.656^{1.05} &\approx 8.477 \end{aligned}$$

Now substitute this value into the effort equation:

$$\begin{aligned} E &\approx 2.4 \times 8.477 \times 1.155 \\ &\approx 20.3448 \times 1.155 \\ &\approx 23.50 \text{ person-months} \end{aligned}$$

Therefore, the preliminary COCOMO effort estimate is:

$$\boxed{E \approx 23.50 \text{ person-months}}$$

2.10.4 Development Time and Average Staffing

Although the assignment only asks for effort E, development time and average staffing help interpret the result. The course development-time equation is:

$$D = cE^d$$

For an organic project, $c = 2.5$ and $d = 0.38$.

$$\begin{aligned} D &= 2.5 \times 23.50^{0.38} \\ 23.50^{0.38} &= e^{0.38 \ln(23.50)} \\ \ln(23.50) &\approx 3.1570 \\ 0.38 \ln(23.50) &\approx 0.38 \times 3.1570 = 1.1997 \\ e^{1.1997} &\approx 3.319 \\ D &\approx 2.5 \times 3.319 = 8.30 \text{ months} \end{aligned}$$

Average people required:

$$\begin{aligned} P &= \frac{E}{D} \\ &= \frac{23.50}{8.30} \\ &\approx 2.83 \text{ people} \end{aligned}$$

Table 12: COCOMO summary

Item	**Estimate**
Function points	264 UFP
Estimated code size	7.656 KLOC
COCOMO mode	Organic
Effort E	23.50 person-months
Development time D	8.30 months
Average staffing P	2.83 people

3 Risk Analysis

3.1 Risk Analysis Scope

The risk analysis selects five important requirements from the previous quality requirements work. The selected requirements cover the success of the student booking lifecycle and the administrator controls that protect that lifecycle.

The five requirements and their weights are based on the relative value percentages used in the previous prioritization work. Since this section considers only five of the seven previous requirements, the five selected values are normalized to sum to 1.0.

Table 13: Selected requirements and normalized weights

Label	Requirement	Previous Value	Normalized Weight
PER-1	Search response time: p95 availability search response within 2.0 seconds.	0.128	0.128/0.884 = 0.145
REL-1	Availability: Carrel is at least 99.5 percent available during the academic term.	0.163	0.163/0.884 = 0.184
INT-1	Integrity: no conflicting active bookings are ever persisted.	0.276	0.276/0.884 = 0.312
ROB-1	Robustness: booking succeeds even if Calendar Service is temporarily unavailable.	0.100	0.100/0.884 = 0.113
SEC-1	Security: identity-service authentication and role enforcement protect every function.	0.217	0.217/0.884 = 0.245
Total		**0.884**	**1.000**

3.2 Risk Definitions

The five risks are system-specific risks that could prevent the selected requirements from being satisfied.

Table 14: Risks considered

Risk	Name	Description	Likelihood
R-A	Peak-load query degradation	Search indexes, caching, or query design are insufficient when many students search after class or before exams.	0.35
R-B	Identity service or role mapping failure	University identity data, session state, or role mapping is unavailable or incorrect.	0.25
R-C	Non-atomic booking transaction bug	Two reservation requests are accepted for the same study space and overlapping interval.	0.20
R-D	Calendar or notification outage	External calendar or notification service is unavailable and blocks or delays booking workflow.	0.30
R-E	Scheduler/time-window bug	Check-in deadlines, no-show release, or time-zone logic behaves incorrectly.	0.25

3.3 Risk Consequences Table

The formulas used are:

$$LossOfObjective(req) = weight(req) \times \sum_{risk} (likelihood(risk) \times impact(risk, req))$$

$$RiskCriticality(risk) = likelihood(risk) \times \sum_{req} (impact(risk, req) \times weight(req))$$

The impact values range from 0 to 1, where 0 means the risk does not meaningfully threaten the requirement and 1 means the risk could fully prevent that requirement from being satisfied.

Table 15: Risk consequences table

Requirement	Description	Weight	R-A	R-B	R-C	R-D	R-E	Loss of Objective
Likelihood			0.35	0.25	0.20	0.30	0.25	
PER-1	Search response time within target	0.145	0.90	0.20	0.40	0.20	0.20	0.0805
REL-1	Monthly availability within target	0.184	0.40	0.80	0.50	0.60	0.50	0.1371
INT-1	No conflicting active bookings	0.312	0.20	0.30	1.00	0.20	0.80	0.1888
ROB-1	Calendar outage tolerance	0.113	0.20	0.20	0.20	0.90	0.30	0.0571
SEC-1	Authentication and role enforcement	0.245	0.10	1.00	0.40	0.20	0.30	0.1225
		Risk Criticality	**0.1098**	**0.1344**	**0.1165**	**0.1058**	**0.1195**	**0.5859 total**

3.4 Detailed Loss of Objective Calculations

The table above is computed row by row. Each row first multiplies each risk likelihood by its impact on that requirement. Then the sum is multiplied by the requirement weight.

PER-1

$$\begin{aligned} Loss(PER\text{-}1) &= 0.145[(0.35)(0.90) + (0.25)(0.20) + (0.20)(0.40) + (0.30)(0.20) + (0.25)(0.20)] \\ &= 0.145[0.315 + 0.050 + 0.080 + 0.060 + 0.050] \\ &= 0.145[0.555] \\ &= 0.0805 \end{aligned}$$

REL-1

$$\begin{aligned} Loss(REL\text{-}1) &= 0.184[(0.35)(0.40) + (0.25)(0.80) + (0.20)(0.50) + (0.30)(0.60) + (0.25)(0.50)] \\ &= 0.184[0.140 + 0.200 + 0.100 + 0.180 + 0.125] \\ &= 0.184[0.745] \\ &= 0.1371 \end{aligned}$$

INT-1

$$\begin{aligned} Loss(INT\text{-}1) &= 0.312[(0.35)(0.20) + (0.25)(0.30) + (0.20)(1.00) + (0.30)(0.20) + (0.25)(0.80)] \\ &= 0.312[0.070 + 0.075 + 0.200 + 0.060 + 0.200] \\ &= 0.312[0.605] \\ &= 0.1888 \end{aligned}$$

ROB-1

$$\begin{aligned} Loss(ROB\text{-}1) &= 0.113[(0.35)(0.20) + (0.25)(0.20) + (0.20)(0.20) + (0.30)(0.90) + (0.25)(0.30)] \\ &= 0.113[0.070 + 0.050 + 0.040 + 0.270 + 0.075] \\ &= 0.113[0.505] \\ &= 0.0571 \end{aligned}$$

SEC-1

$$\begin{aligned} Loss(SEC\text{-}1) &= 0.245[(0.35)(0.10) + (0.25)(1.00) + (0.20)(0.40) + (0.30)(0.20) + (0.25)(0.30)] \\ &= 0.245[0.035 + 0.250 + 0.080 + 0.060 + 0.075] \\ &= 0.245[0.500] \\ &= 0.1225 \end{aligned}$$

3.5 Detailed Risk Criticality Calculations

Risk criticality is computed column by column. Each risk criticality multiplies the risk likelihood by the weighted sum of the risk's impacts on all five requirements.

R-A: Peak-load query degradation

$$\begin{aligned} \mathit{Criticality}(R\text{-}A) &= 0.35[(0.90)(0.145) + (0.40)(0.184) + (0.20)(0.312) + (0.20)(0.113) + (0.10)(0.245)] \\ &= 0.35[0.1305 + 0.0736 + 0.0624 + 0.0226 + 0.0245] \\ &= 0.35[0.3136] \\ &= 0.1098 \end{aligned}$$

R-B: Identity service or role mapping failure

$$\begin{aligned} \mathit{Criticality}(R\text{-}B) &= 0.25[(0.20)(0.145) + (0.80)(0.184) + (0.30)(0.312) + (0.20)(0.113) + (1.00)(0.245)] \\ &= 0.25[0.0290 + 0.1472 + 0.0936 + 0.0226 + 0.2450] \\ &= 0.25[0.5374] \\ &= 0.1344 \end{aligned}$$

R-C: Non-atomic booking transaction bug

$$\begin{aligned} \mathit{Criticality}(R\text{-}C) &= 0.20[(0.40)(0.145) + (0.50)(0.184) + (1.00)(0.312) + (0.20)(0.113) + (0.40)(0.245)] \\ &= 0.20[0.0580 + 0.0920 + 0.3120 + 0.0226 + 0.0980] \\ &= 0.20[0.5826] \\ &= 0.1165 \end{aligned}$$

R-D: Calendar or notification outage

$$\begin{aligned} \mathit{Criticality}(R\text{-}D) &= 0.30[(0.20)(0.145) + (0.60)(0.184) + (0.20)(0.312) + (0.90)(0.113) + (0.20)(0.245)] \\ &= 0.30[0.0290 + 0.1104 + 0.0624 + 0.1017 + 0.0490] \\ &= 0.30[0.3525] \\ &= 0.1058 \end{aligned}$$

R-E: Scheduler/time-window bug

$$\begin{aligned} \mathit{Criticality}(R\text{-}E) &= 0.25[(0.20)(0.145) + (0.50)(0.184) + (0.80)(0.312) + (0.30)(0.113) + (0.30)(0.245)] \\ &= 0.25[0.0290 + 0.0920 + 0.2496 + 0.0339 + 0.0735] \\ &= 0.25[0.4780] \\ &= 0.1195 \end{aligned}$$

The largest risk criticality is **R-B identity service or role mapping failure** at 0.1344. The second largest is **R-E scheduler/time-window bug** at 0.1195, followed closely by **R-C non-atomic booking transaction bug** at 0.1165. These are the tall poles in the risk table and should receive early mitigation attention.

3.6 Risk Countermeasures

The countermeasure table estimates how much each countermeasure reduces each risk. A reduction of 0 means no meaningful risk reduction; a reduction of 1 means that the countermeasure would fully remove that risk.

The formulas used are:

$$EffectOfCountermeasure(cm) = \sum_{risk}(reduction(cm, risk) \times criticality(risk))$$

$$CombinedRiskReduction(risk) = 1 - \prod_{cm}(1 - reduction(cm, risk))$$

Table 16: Risk countermeasures table

Countermeasure	Description	R-A	R-B	R-C	R-D	R-E	Effect of Countermeasure
Risk criticality		0.1098	0.1344	0.1165	0.1058	0.1195	
CM-1	Add query indexes, caching, and a load-test gate for search before release.	0.75	0.05	0.10	0.10	0.10	0.1232
CM-2	Add identity-service fallback handling, short-lived role cache, and role mapping tests.	0.05	0.80	0.05	0.05	0.05	0.1301
CM-3	Use database transactions, unique overlap checks, and concurrency tests for reservation.	0.10	0.10	0.85	0.10	0.25	0.1639
CM-4	Use asynchronous notification/calendar queue with retry and fallback confirmation.	0.10	0.10	0.05	0.80	0.15	0.1328
CM-5	Add scheduler simulations, time-zone tests, and no-show release audit logging.	0.05	0.05	0.10	0.10	0.75	0.1241
	Combined Risk Reduction	**0.8172**	**0.8538**	**0.8903**	**0.8615**	**0.8637**	

3.7 Detailed Effect of Countermeasure Calculations

CM-1

$$\begin{aligned} Effect(CM\text{-}1) &= (0.75)(0.1098) + (0.05)(0.1344) + (0.10)(0.1165) + (0.10)(0.1058) + (0.10)(0.1195) \\ &= 0.0824 + 0.0067 + 0.0117 + 0.0106 + 0.0120 \\ &= 0.1232 \end{aligned}$$

CM-2

$$\begin{aligned} Effect(CM\text{-}2) &= (0.05)(0.1098) + (0.80)(0.1344) + (0.05)(0.1165) + (0.05)(0.1058) + (0.05)(0.1195) \\ &= 0.0055 + 0.1075 + 0.0058 + 0.0053 + 0.0060 \\ &= 0.1301 \end{aligned}$$

CM-3

$$\begin{aligned} Effect(CM\text{-}3) &= (0.10)(0.1098) + (0.10)(0.1344) + (0.85)(0.1165) + (0.10)(0.1058) + (0.25)(0.1195) \\ &= 0.0110 + 0.0134 + 0.0990 + 0.0106 + 0.0299 \\ &= 0.1639 \end{aligned}$$

CM-4

$$\begin{aligned} Effect(CM\text{-}4) &= (0.10)(0.1098) + (0.10)(0.1344) + (0.05)(0.1165) + (0.80)(0.1058) + (0.15)(0.1195) \\ &= 0.0110 + 0.0134 + 0.0058 + 0.0846 + 0.0179 \\ &= 0.1328 \end{aligned}$$

CM-5

$$\begin{aligned} Effect(CM\text{-}5) &= (0.05)(0.1098) + (0.05)(0.1344) + (0.10)(0.1165) + (0.10)(0.1058) + (0.75)(0.1195) \\ &= 0.0055 + 0.0067 + 0.0117 + 0.0106 + 0.0896 \\ &= 0.1241 \end{aligned}$$

CM-3 has the largest effect of countermeasure because the non-atomic booking transaction risk threatens the highest-weight requirement, INT-1. CM-4 and CM-2 are also important because calendar/notification resilience and authentication correctness affect several high-value requirements.

3.8 Detailed Combined Risk Reduction Calculations

Combined risk reduction assumes the countermeasures reduce a risk independently. Therefore, the remaining unreduced portion is the product of all remaining portions $(1 - reduction)$.

R-A

$$\begin{aligned} CombinedReduction(R\text{-}A) &= 1 - (1 - 0.75)(1 - 0.05)(1 - 0.10)(1 - 0.10)(1 - 0.05) \\ &= 1 - (0.25)(0.95)(0.90)(0.90)(0.95) \\ &= 1 - 0.1828 \\ &= 0.8172 \end{aligned}$$

R-B

$$\begin{aligned} CombinedReduction(R\text{-}B) &= 1 - (1 - 0.05)(1 - 0.80)(1 - 0.10)(1 - 0.10)(1 - 0.05) \\ &= 1 - (0.95)(0.20)(0.90)(0.90)(0.95) \\ &= 1 - 0.1462 \\ &= 0.8538 \end{aligned}$$

R-C

$$\begin{aligned} CombinedReduction(R\text{-}C) &= 1 - (1 - 0.10)(1 - 0.05)(1 - 0.85)(1 - 0.05)(1 - 0.10) \\ &= 1 - (0.90)(0.95)(0.15)(0.95)(0.90) \\ &= 1 - 0.1097 \\ &= 0.8903 \end{aligned}$$

R-D

$$\begin{aligned} CombinedReduction(R\text{-}D) &= 1 - (1 - 0.10)(1 - 0.05)(1 - 0.10)(1 - 0.80)(1 - 0.10) \\ &= 1 - (0.90)(0.95)(0.90)(0.20)(0.90) \\ &= 1 - 0.1385 \\ &= 0.8615 \end{aligned}$$

R-E

$$\begin{aligned} CombinedReduction(R\text{-}E) &= 1 - (1 - 0.10)(1 - 0.05)(1 - 0.25)(1 - 0.15)(1 - 0.75) \\ &= 1 - (0.90)(0.95)(0.75)(0.85)(0.25) \\ &= 1 - 0.1363 \\ &= 0.8637 \end{aligned}$$

3.9 Residual Risk After Countermeasures

Residual criticality is computed as:

$$ResidualCriticality(risk) = RiskCriticality(risk) \times (1 - CombinedRiskReduction(risk))$$

Table 17: Residual risk after countermeasures

Risk	Original Criticality	Combined Reduction	Residual Criticality
R-A	0.1098	0.8172	$0.1098(1 - 0.8172) = 0.0201$
R-B	0.1344	0.8538	$0.1344(1 - 0.8538) = 0.0196$
R-C	0.1165	0.8903	$0.1165(1 - 0.8903) = 0.0128$
R-D	0.1058	0.8615	$0.1058(1 - 0.8615) = 0.0146$
R-E	0.1195	0.8637	$0.1195(1 - 0.8637) = 0.0163$
Total	**0.5859**		**0.0834**

The countermeasures reduce the total risk criticality from 0.5859 to 0.0834. The overall percentage reduction is:

$$\begin{aligned} OverallReduction &= 1 - \frac{0.0834}{0.5859} \\ &= 1 - 0.1424 \\ &= 0.8576 \\ &= 85.76\% \end{aligned}$$

4 LTL Quality Attributes and Long-Term Behaviour

4.1 LTL Notation

The following notation is used:

Table 18: LTL notation

Symbol	Name	Meaning
$\Box p$	Henceforth / Always	p must be true in every state from now on.
$\Diamond p$	Eventually	p must become true in some future state.
$\bigcirc p$	Next	p must be true in the next state.
$p \,\mathcal{U}\, q$	Until	p must remain true until q eventually becomes true.
$p \,\mathcal{W}\, q$	Unless / Weak Until	p must remain true until q becomes true, but q is not guaranteed to happen.

4.2 Variables and Predicates

The variables and predicates below are derived from the Carrel domain model, functional model, and state diagram. They define the vocabulary used in the LTL statements.

Table 19: Variables and predicates used in LTL constraints

Variable / Predicate	**Meaning**
u	A user of Carrel.
s	A student user.
sp	A study space.
b	A booking.
d	A dispute.
p	A pull request or implementation change.
$authenticated(u)$	User u has an active authenticated Carrel session.
$authOK(u)$	The University Identity Service has accepted the user's credentials.
$roleValid(u)$	Carrel has loaded a valid role for user u.
$admin(u)$	User u has the Administrator role.
$dashboardLoaded(u)$	The correct dashboard for u has been loaded.
$reserveReq(s, sp)$	Student s submits a reservation request for study space sp.
$rulesOK(s, sp)$	All applicable booking policies are satisfied for the request.
$spaceAvailable(sp)$	The study space is available for the requested interval.
$bookingConfirmed(b)$	Booking b has been confirmed.
$bookingRejected(s, sp)$	The reservation request is rejected with a visible explanation.
$activeBooking(b)$	Booking b is Confirmed, AwaitingCheckIn, or CheckedIn.
$conflict(b)$	Booking b overlaps another active booking or active maintenance closure.
$cancelled(b)$	Booking b has been cancelled.
$completed(b)$	Booking b has reached its natural completed state.
$checkInWindowOpen(b)$	Booking b is inside its check-in window.
$submitCheckIn(s, b)$	Student s submits a check-in code or scans a QR code for booking b.
$validCheckInCode(b)$	The submitted check-in code belongs to b and has not been used.
$beforeCheckInDeadline(b)$	The current state is before or at b's check-in deadline.
$checkedIn(b)$	Booking b has a successful CheckIn record.
$deadlinePassed(b)$	The check-in deadline for booking b has passed.
$noCheckIn(b)$	Booking b has no CheckIn record.
$noShowReleased(b)$	Booking b has been released due to no-show.
$strikeActive(s)$	Student s has an active no-show strike.
$cancelReq(s, b)$	Student s requests to cancel booking b.
$beforeCancelDeadline(b)$	Booking b is still inside the allowed cancellation window.
$cancellationNoticeSent(s, b)$	A cancellation notice has been sent to student s for booking b.
$calendarUnavailable$	The Calendar Service is unavailable.
$calendarRecovered$	The Calendar Service has recovered.
$bookingConfirmationSent(b)$	The booking confirmation was sent even if calendar sync is unavailable.
$inviteDelivered(b)$	The CalendarInvite for booking b has been delivered or synchronized.
$disputeSubmitted(d)$	Dispute d has been submitted.
$underReview(d)$	Dispute d is under administrator review.
$disputeResolved(d)$	Dispute d has a recorded resolution.

Variable / Predicate	Meaning
$disputeNoticeSent(d)$	A dispute update notice has been sent.
$policyUpdateReq(u)$	User u attempts to update booking policy/rules.
$overrideReq(u)$	User u attempts to apply an administrative override.
$closeSpaceReq(u)$	User u attempts to create or update a maintenance closure.
$prOpened(p)$	An implementation pull request or change p has been opened.
$reviewed(p)$	Pull request p has been reviewed.
$merged(p)$	Pull request p has been merged.
$rejected(p)$	Pull request p has been rejected.
$testsPassed(p)$	Pull request p passed the required automated tests.
$storedInCanada(b)$	Booking data for b is stored in an approved Canadian data location.

4.3 Ten English Attributes and LTL Statements

Each item is first written in natural language and then using LTL. The statements are intended to describe long-term behavior over executions of the Carrel system.

Table 20: Quality attributes and LTL constraints

ID	Attribute	English Description	LTL Statement
LTL-1	Security / usability	Whenever a user's credentials are accepted and a valid role is loaded, the system eventually loads the correct dashboard for that user.	$\square((authOK(u) \land roleValid(u)) \rightarrow \lozenge dashboardLoaded(u))$
LTL-2	Security	A user may request reservation, check-in, cancellation, or dispute submission only while authenticated.	$\square((reserveReq(s, sp) \lor submitCheckIn(s, b) \lor cancelReq(s, b) \lor disputeSubmitted(d)) \rightarrow authenticated(u))$
LTL-3	Performance / functional success	If a student requests an available space and all booking rules are satisfied, the request eventually results in a confirmed booking.	$\square((reserveReq(s, sp) \land spaceAvailable(sp) \land rulesOK(s, sp)) \rightarrow \lozenge bookingConfirmed(b))$
LTL-4	Integrity	Once a booking is active, it must remain non-conflicting unless it is cancelled, released as a no-show, or completed.	$\square(activeBooking(b) \rightarrow (\neg conflict(b) \ \mathcal{W} \ (cancelled(b) \lor noShowReleased(b) \lor completed(b))))$
LTL-5	Integrity / check-in correctness	If a student submits a valid check-in code before the deadline while the check-in window is open, the booking is checked in in the next system state.	$\square((checkInWindowOpen(b) \land submitCheckIn(s, b) \land validCheckInCode(b) \land beforeCheckInDeadline(b)) \rightarrow \bigcirc checkedIn(b))$

ID	Attribute	English Description	LTL Statement
LTL-6	Reliability / no-show enforcement	If an active booking passes its check-in deadline without a check-in, then it is eventually released and the student eventually receives an active no-show strike.	$\square((activeBooking(b) \land deadlinePassed(b) \land noCheckIn(b)) \rightarrow \Diamond(noShowReleased(b) \land strikeActive(s)))$
LTL-7	Functional correctness / notification	If a student cancels inside the allowed cancellation window, then the booking eventually becomes cancelled and a cancellation notice is eventually sent.	$\square((cancelReq(s,b) \land beforeCancelDeadline(b)) \rightarrow \Diamond(cancelled(b) \land cancellationNoticeSent(s,b)))$
LTL-8	Robustness	If the calendar service is unavailable after a booking is confirmed, the booking confirmation must still eventually be sent, and the calendar outage persists only until the service recovers or the invite is delivered.	$\square((bookingConfirmed(b) \land calendarUnavailable) \rightarrow (\Diamond bookingConfirmationSent(b) \land (calendarUnavailable\ \mathcal{U}\ (calendarRecovered \lor inviteDelivered(b)))))$
LTL-9	Security / role control	Only administrators may update booking policies, apply overrides, or create maintenance closures.	$\square((policyUpdateReq(u) \lor overrideReq(u) \lor closeSpaceReq(u)) \rightarrow admin(u))$
LTL-10	Process / maintainability	Every implementation change must eventually be reviewed, and after review it must either pass tests and be merged or be rejected.	$\square(prOpened(p) \rightarrow \Diamond(reviewed(p) \land ((testsPassed(p) \land merged(p)) \lor rejected(p))))$

4.4 Why the LTL Statements Match Carrel

The first group of LTL statements protects user authentication and normal navigation. LTL-1 and LTL-2 connect the unauthenticated and authenticated parts of the state diagram to the domain model's user and role concepts. LTL-3 through LTL-7 describe the long-term booking lifecycle: reservation, active booking integrity, check-in, no-show release, and cancellation. LTL-8 captures robustness when Calendar Service is unavailable; the key point is that booking confirmation should not be blocked permanently by an external outage. LTL-9 expresses administrator-only control for functions that could affect many users. LTL-10 is a process requirement that protects long-term maintainability by requiring review and automated tests for implementation changes.

5 Final Summary

The preliminary effort estimate for Carrel is **264 function points**, which converts to **7,656 SLOC** or **7.656 KLOC** using the selected high-level language proxy. Using organic COCOMO with the documented project attribute multiplier gives an estimated effort of **23.50 person-months**, approximately **8.30 months** with an average of **2.83 people**.

The risk analysis identifies five major risks. The highest original risk criticality is the identity service or role mapping failure risk. After the proposed countermeasures, the total risk criticality is reduced from **0.5859** to **0.0834**, an estimated reduction of **85.76 percent**. Finally, the ten LTL statements express Carrel's long-term security, performance, integrity, reliability, robustness, and process behavior in a form that can be checked against the domain model and behavioral model.